- RAG Learning

Differentiation

Differentiation from First Principles

Definition: The derivative of $f(x)$ from first principles is:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

This gives the gradient of the tangent to the curve at any point.

Worked Example: First Principles for $f(x) = x^3$

Solution:

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^3 - x^3}{h}$$

Expand $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$:

$$= \lim_{h \to 0} \frac{3x^2h + 3xh^2 + h^3}{h} = \lim_{h \to 0}(3x^2 + 3xh + h^2) = 3x^2$$

Standard Derivatives

y or $f(x)$	$\frac{dy}{dx}$ or $f'(x)$	Notes

x^n	nx^{n-1}	Power rule (all rational n)
e^x	e^x	Self-derivative
e^{kx}	ke^{kx}	Chain rule applied
a^x	$a^x \ln a$	General exponential
$\ln x$	$\frac{1}{x}$	$x > 0$
$\sin x$	$\cos x$	x in radians
$\cos x$	$-\sin x$	x in radians
$\tan x$	$\sec^2 x$	x in radians
$\sin(kx)$	$k\cos(kx)$	Chain rule applied
$\cos(kx)$	$-k\sin(kx)$	Chain rule applied

Differentiation Rules

Chain Rule

Chain Rule: If $y = f(g(x))$, then:

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

Or equivalently: if $y = f(u)$ and $u = g(x)$, then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$.

Worked Example: Chain Rule

Differentiate $y = (3x^2 + 1)^5$.

Solution: Let $u = 3x^2 + 1$, so $y = u^5$.

$$\frac{dy}{dx} = 5u^4 \cdot 6x = 30x(3x^2+1)^4$$

Product Rule

Product Rule: If $y = uv$, then:

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Worked Example: Product Rule

Differentiate $y = x^2e^{3x}$.

Solution: Let $u = x^2$ and $v = e^{3x}$.

$$\frac{dy}{dx} = x^2 \cdot 3e^{3x} + e^{3x} \cdot 2x = e^{3x}(3x^2+2x) = xe^{3x}(3x+2)$$

Quotient Rule

Quotient Rule: If $y = \frac{u}{v}$, then:

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Worked Example: Quotient Rule

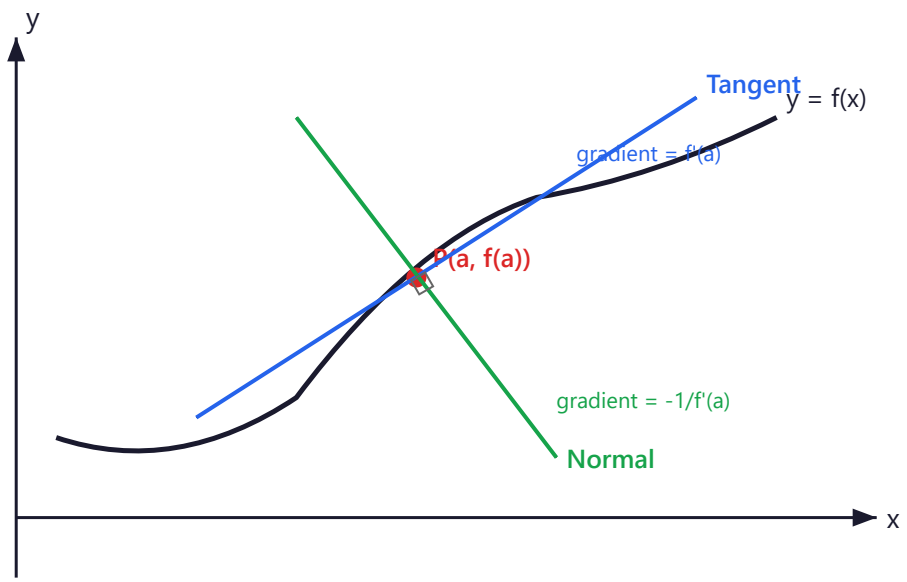
Differentiate $y = \dfrac{\sin x}{x^2}$.

Solution: Let $u = \sin x$, $v = x^2$:

$$\frac{dy}{dx} = \frac{x^2 \cos x - \sin x \cdot 2x}{x^4} = \frac{x \cos x - 2 \sin x}{x^3}$$

Exam Tip: Remember the quotient rule with the mnemonic "low d-high minus high d-low, over low squared". Always simplify your final answer by factorising where possible.

Tangents and Normals



At the point $(a, f(a))$:

- **Tangent line:** $y - f(a) = f'(a)(x - a)$
- **Normal line:** $y - f(a) = -\dfrac{1}{f'(a)}(x - a)$ (perpendicular to tangent)

Implicit Differentiation

Implicit Differentiation: When y is defined implicitly by an equation (not written as $y = \ldots$), differentiate each term with respect to x, applying the chain rule to terms involving y:

$$\frac{d}{dx}[f(y)] = f'(y) \cdot \frac{dy}{dx}$$

Worked Example: Implicit Differentiation

Find $\frac{dy}{dx}$ given $x^2 + y^2 = 25$.

Solution: Differentiate both sides with respect to x:

$$2x + 2y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

Worked Example: Harder Implicit Differentiation

Find $\frac{dy}{dx}$ given $x^2y + xy^3 = 6$.

Solution: Use the product rule on each term:

$$\left(2xy + x^2\frac{dy}{dx}\right) + \left(y^3 + 3xy^2\frac{dy}{dx}\right) = 0$$

$$\frac{dy}{dx}(x^2 + 3xy^2) = -(2xy + y^3)$$

$$\frac{dy}{dx} = \frac{-(2xy + y^3)}{x^2 + 3xy^2}$$

Stationary Points

Definition: A stationary point occurs where $f'(x) = 0$. The nature of the stationary point is determined by the second derivative or by examining the sign of the first derivative either side of the point.

Second Derivative Test	Nature
$f''(a) > 0$	Local minimum at $x = a$
$f''(a) < 0$	Local maximum at $x = a$

$f''(a) = 0$	Inconclusive — check sign of $f'(x)$ either side

Worked Example: Finding Stationary Points

Find the stationary points of $f(x) = 2x^3 - 9x^2 + 12x - 4$ and determine their nature.

Solution:

$$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x-1)(x-2)$$

Setting $f'(x) = 0$: $x = 1$ or $x = 2$.

$$f''(x) = 12x - 18$$

At $x = 1$: $f''(1) = -6 < 0$ — local maximum. $f(1) = 1$, so max at $(1, 1)$.

At $x = 2$: $f''(2) = 6 > 0$ — local minimum. $f(2) = 0$, so min at $(2, 0)$.

Connected Rates of Change

Connected Rates of Change: Use the chain rule to relate rates:

$$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$$

This connects the rate of change of one variable to another when both depend on time.

Worked Example: Connected Rates

A spherical balloon is being inflated so that its volume increases at $20 \text{ cm}^3/\text{s}$. Find the rate of increase of the radius when $r = 5$ cm.

Solution: $V = \frac{4}{3}\pi r^3$, so $\frac{dV}{dr} = 4\pi r^2$.

$$\frac{dr}{dt} = \frac{dV}{dt} \cdot \frac{dr}{dV} = 20 \cdot \frac{1}{4\pi r^2}$$

When $r = 5$: $\frac{dr}{dt} = \frac{20}{4\pi(25)} = \frac{1}{5\pi} \approx 0.0637$ cm/s.

Parametric Differentiation

Parametric Differentiation: If $x = f(t)$ and $y = g(t)$, then:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Worked Example: Parametric Curve

A curve is defined by $x = t^2 + 1$, $y = t^3 - t$. Find $\frac{dy}{dx}$ and the equation of the tangent when $t = 2$.

Solution:

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = 3t^2 - 1$$

$$\frac{dy}{dx} = \frac{3t^2 - 1}{2t}$$

When $t = 2$: $\frac{dy}{dx} = \frac{11}{4}$, and the point is $(5, 6)$.

$$y - 6 = \frac{11}{4}(x - 5) \implies y = \frac{11}{4}x - \frac{31}{4}$$

Exam Tip: For second parametric derivatives, use:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

Do NOT simply differentiate $\frac{dy}{dx}$ with respect to t and forget to divide by $\frac{dx}{dt}$.

Increasing and Decreasing Functions

- f is **increasing** on an interval if $f'(x) \geq 0$ for all x in that interval
- f is **decreasing** on an interval if $f'(x) \leq 0$ for all x in that interval

Worked Example: Finding Increasing/Decreasing Intervals

Find the intervals where $f(x) = x^3 - 6x^2 + 9x + 1$ is increasing and decreasing.

Solution:

$$f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3)$$

Interval	Sign of $f'(x)$	Behaviour

$x < 1$	$+$	Increasing
$1 < x < 3$	$-$	Decreasing
$x > 3$	$+$	Increasing

Exam Tip: To "show that f is increasing for all x", prove that $f'(x) \geq 0$ for all x. A common approach is to complete the square in $f'(x)$ and show the result is always non-negative.

Points of Inflection

Point of Inflection: A point where the curve changes concavity (from concave up to concave down, or vice versa). Necessary condition: $f''(x) = 0$. The sign of $f''(x)$ must change either side to confirm it.

A **stationary point of inflection** has both $f'(x) = 0$ and $f''(x) = 0$ at that point, with a sign change in f'' confirming it is not a maximum or minimum.

At point $x = a$	Nature
$f'(a) = 0,\ f''(a) < 0$	Local maximum
$f'(a) = 0,\ f''(a) > 0$	Local minimum
$f'(a) = 0,\ f''(a) = 0$, sign change in f''	Stationary point of inflection
$f'(a) \neq 0,\ f''(a) = 0$, sign change in f''	Non-stationary point of inflection

Worked Example: Finding Points of Inflection

Find any points of inflection of $f(x) = x^4 - 4x^3$.

Solution:

$$f''(x) = 12x^2 - 24x = 12x(x-2)$$

Setting $f''(x) = 0$: $x = 0$ or $x = 2$.

- At $x = 0$: f'' changes from $+$ to $-$ — point of inflection at $(0,\ 0)$.
- At $x = 2$: f'' changes from $-$ to $+$ — point of inflection at $(2,\ -16)$.

Exam Tip: $f''(a) = 0$ alone does NOT confirm a point of inflection — you must verify the sign change. For example, $f(x) = x^4$ has $f''(0) = 0$ but f'' does not change sign at 0, so it is a minimum, not an inflection.